

Math Module

Python Maths

Math Module

Python has a built-in module that you can use for mathematical tasks.

The math module has a set of methods and constants.

Import math

- # Import math Library
`import math`

math.cos

- # Import math Library
`import math`

Return the cosine of different numbers
`print (math.cos(0.00))`

- #Import math Library
`import math`

#Return factorial of a number
`print(math.factorial(9))`

- # Import math Library
`import math`

Check whether some values are NaN or not

```
print (math.isnan (56))  
print (math.isnan (-45.34))  
print (math.isnan (+45.34))  
print (math.isnan (math.inf))  
print (math.isnan (float("nan")))  
print (math.isnan (float("inf")))  
print (math.isnan (float("-inf")))  
print (math.isnan (math.nan))
```

- # Import math Library
`import math`

```
# Return the natural logarithm of different numbers  
print(math.log(2.7183))  
print(math.log(2))  
print(math.log(1))
```

- # Import math Library
`import math`

#Return the value of 9 raised to the power of 3
`print(math.pow(9, 3))`

- # Import math Library
`import math`

Return the square root of different numbers
`print (math.sqrt(9))`

- # Import math library
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Round numbers down to the nearest integer
`print(math.floor(0.6))`
`print(math.floor(1.4))`

- # Import math library
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Round a number upward to its nearest integer
`print(math.ceil(1.4))`
`print(math.ceil(5.3))`

Solving equations in Python

- Solving equations in Python can be done using various libraries, such as sympy for symbolic mathematics or numpy for numerical solutions.

```
import numpy as np
```

```
# Define the coefficients of the equations
```

```
A = np.array([[3, 1], [1, 2]])
```

```
B = np.array([9, 8])
```

```
# Solve the system of equations
```

```
solutions = np.linalg.solve(A, B)
```

```
print(solutions)
```

```
import numpy as np
```

```
# Coefficients of the equations:
```

```
#  $x + y + z = 6$ 
```

```
#  $2x + 3y + z = 14$ 
```

```
#  $4x + y + 2z = 20$ 
```

```
A = np.array([[1, 1, 1], [2, 3, 1], [4, 1, 2]])
```

```
B = np.array([6, 14, 20])
```

```
# Solve the system of equations
```

```
solutions = np.linalg.solve(A, B)
```

```
print(solutions)
```

Basic Probability Calculations

```
import numpy as np
```

```
# Probability of an event
```

```
prob_event = 0.3
```

```
print("Probability of the event:", prob_event)
```

```
# Complementary probability
```

```
prob_not_event = 1 - prob_event
```

```
print("Probability of not the event:", prob_not_event)
```

```
import numpy as np
import matplotlib.pyplot as plt
# Generate samples
uniform_samples = np.random.uniform(0, 1, 1000)
normal_samples = np.random.normal(0, 1, 1000)
binomial_samples = np.random.binomial(10, 0.5, 1000)
poisson_samples = np.random.poisson(5, 1000)
# Plot samples
plt.figure(figsize=(10, 8))
plt.subplot(2, 2, 1)
plt.hist(uniform_samples, bins=30, color='blue', alpha=0.7)
plt.title('Uniform Distribution')
plt.subplot(2, 2, 2)
plt.hist(normal_samples, bins=30, color='green', alpha=0.7)
plt.title('Normal Distribution')
plt.subplot(2, 2, 3)
plt.hist(binomial_samples, bins=30, color='red', alpha=0.7)
plt.title('Binomial Distribution')
plt.subplot(2, 2, 4)
plt.hist(poisson_samples, bins=30, color='purple', alpha=0.7)
plt.title('Poisson Distribution')
plt.tight_layout()
plt.show()
```


Resources

https://www.w3schools.com/python/module_math.asp
